

Backtesting Probability of Default Models

From Homogeneous Grades to Heterogeneous Portfolios

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Internal Ratings-Based Modeling Process

- The development of an Internal Ratings-Based (IRB) model consists of two main areas: the development of the risk differentiation function and risk quantification.
- Development of the risk differentiation function is the first step in the overall model development process. The main goal of this step is to ensure the model can adequately rank risk, providing meaningful differentiation between low- and high-risk obligors or facilities.
- After developing the risk differentiation function, practitioners perform the risk quantification process, which essentially involves assigning an appropriate risk level to the output of the risk differentiation function.
- Risk quantification typically consists of specific steps such as: assignment of the base or best estimate, quantification of appropriate adjustments (AA), quantification of the margin of conservatism (MoC), and quantification of any other additional conservatism, for instance that required by certain local regulators. The first two steps are typically referred to as calibration.
- These steps apply to all three risk parameters: Probability of Default (PD), Loss Given Default (LGD), and Exposure at Default (EAD).

Probability of Default Backtesting

- As risk quantification is a regulatory requirement, practitioners pay significant attention to complying with these requirements.
- In addition to adhering to the prescribed risk quantification process, practitioners are required to assess the appropriateness of model estimates. This is typically referred to as backtesting.
- The primary purpose of backtesting in this context is to ensure that model estimates accurately predict the occurrence of defaults, i.e., that PD estimates provide reliable forecasts of default rates.
- In practice, this is one of the most comprehensive validation exercises, typically investigating model estimates across different portfolio subsegments and time dimensions.
- Regardless of the type of PD model output (discrete or continuous), the model should be backtested across the full range of estimates and at the portfolio and segment levels.

Probability of Default Backtesting cont.

- When moving from rating-grade to portfolio-level testing for discrete model outputs, practitioners often overlook that portfolio (segment or subsegment) data consist of heterogeneous obligors assigned estimates derived from homogeneous rating grades.
- Practitioners typically assume that, at the portfolio level, PD follows a binomial distribution, where PD is the average probability of default across all obligors and the corresponding variance is given by $\frac{PD(1-PD)}{N}$.
- However, given that the primary level of estimation for discrete model outputs is the rating grade, it is reasonable to assume that portfolio-level defaults arise from a series of independent Bernoulli trials that are not necessarily identically distributed. Moreover, considering the regulatory requirement for rating scale heterogeneity, it is evident that rating-grade PDs are not identically distributed.
- Overlooking this fact leads to the inappropriate application of certain statistical tests for the purpose of PD backtesting at the portfolio or subsegment level.
- The Poisson binomial distribution describes the discrete probability distribution of the sum of independent Bernoulli trials that are not necessarily identically distributed.
- The purpose of this presentation is to demonstrate the difference between the current prevalent practice and the proper treatment of portfolio-level information in PD backtesting exercises.

Poisson binomial distribution

- The Poisson binomial distribution describes the discrete probability distribution of the sum of independent Bernoulli trials that are not necessarily identically distributed.
- Unlike the binomial distribution, which assumes homogeneity of the success probability across observations, the Poisson binomial distribution accounts for heterogeneity in the probabilities assigned to the analyzed observations.
- For the purpose of PD backtesting, the concept of the Poisson binomial distribution naturally applies to portfolio-level or subsegment-level testing, given that the output of PD models is typically calibrated at the level of homogeneous rating grades.

Poisson binomial distribution cont.

The following formulas define the main characteristics of the Poisson binomial distribution:

1 Mean:

$$\mu = \mathbb{E}[X] = \sum_{i=1}^n p_i$$

2 Variance:

$$\sigma^2 = \text{Var}(X) = \sum_{i=1}^n p_i(1 - p_i)$$

3 Probability Mass Function(PMF):

$$P(X = k) = \sum_{A \in F_k} \prod_{i \in A} p_i \prod_{j \in A^c} (1 - p_j), \quad k = 0, 1, \dots, n$$

where F_k , denotes the collection of all subsets of $1, \dots, n$ containing exactly k elements and A^c is the complement of A .

Given the above characteristics, the Poisson binomial distribution can be used as a natural extension of the standard binomial test for PD backtesting, while properly accounting for heterogeneity among rating-grade PD estimates.

Simulation Study

The purpose of this simulation study is to demonstrate the impact of assuming homogeneity of estimates at the portfolio (segment or subsegment) level when the model is calibrated at the rating-grade level. Specifically, the simulation consists of two main steps:

- 1 Comparison of the portfolio-level PD distribution under the assumption that PD follows a binomial distribution at the rating-grade level. Additionally, the ratio between the standard deviation of the simulated distribution and the standard deviations implied by the naive binomial and Poisson binomial distributions is provided.
- 2 Assessment of the Type I error of three tests that can be used for portfolio-level PD backtesting: the z-score test, the exact binomial test, and the Poisson binomial test. The first two approaches are typically used in practice, while the third approach is proposed by [Andrija Djurovic](#).

Simulation Study cont.

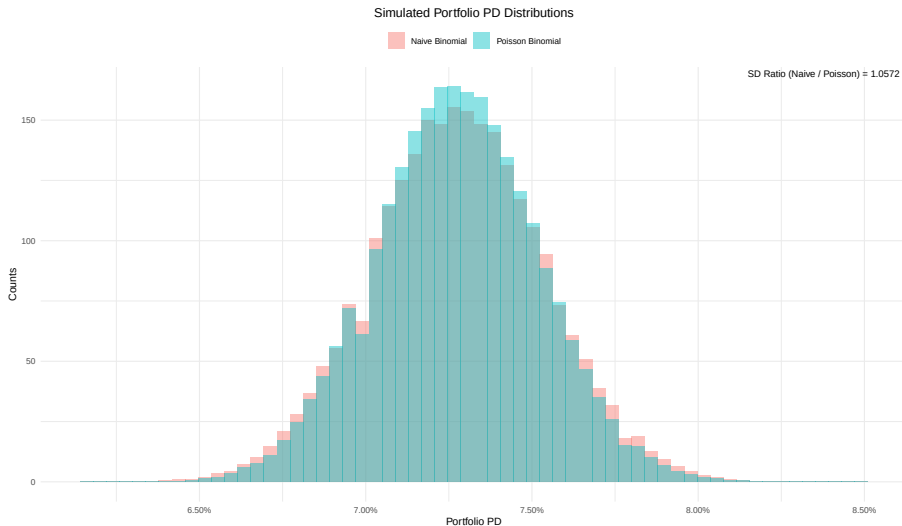
The simulation is based on the following rating scale:

##	rg	n	pd
##	RG1	300	0.005
##	RG2	1000	0.010
##	RG3	2200	0.025
##	RG4	3000	0.040
##	RG5	1800	0.070
##	RG6	1200	0.200
##	RG7	500	0.350

Both simulation steps rely on Monte Carlo simulations, which independently generate defaults from the binomial distribution at the rating-grade level using the provided PD estimates (pd) and number of obligors (n).

The final slide of this presentation illustrates the calculation of the Poisson binomial test p-value, which essentially sums the probabilities (PMF values) associated with default counts that are lower than the observed number of defaults.

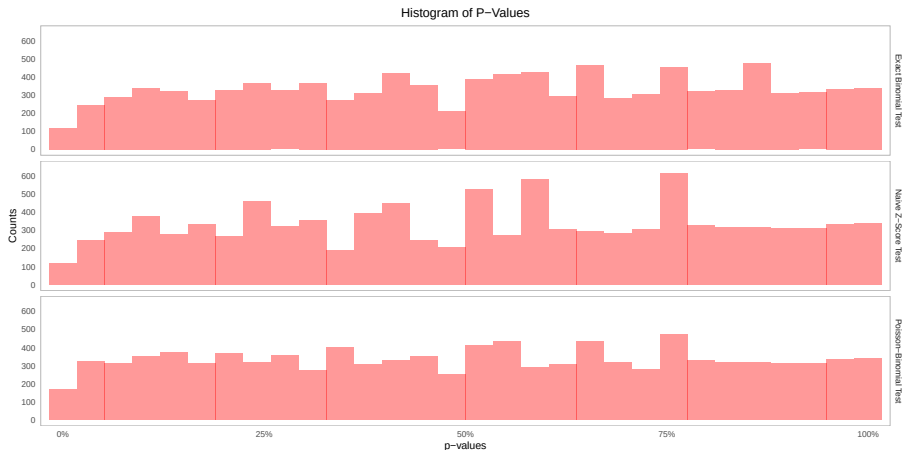
Simulation Results



Simulation Results cont.

Type I error:

##	alpha = 1%	alpha = 5%	alpha = 10%
## Exact Binomial Test	0.66%	3.62%	7.69%
## Naive Z-Score Test	0.68%	3.62%	7.69%
## Poisson-Binomial Test	0.96%	4.73%	9.90%



Simulation Results cont.

